This value introduced into the equation gives the theoretical value of the RSD_{TR}. At this concentration level, RSD_{TR} must usually be close to 4% as:

$$RSD_{TR} = 2^{1-0.5\log_{10}(10^{-2})} = 2^{1-0.5\times(-2)} = 2^{1+1} = 2^2 = 4\%$$

The remarkable success of this proposal led the official control bodies to create an acceptance criterion to check whether a standard deviation of reproducibility is acceptable or not. This is the *HorRat*, defined by the formula (7.22). This is the ratio of an observed relative standard deviation RSD_R calculated at the end of an interlaboratory study, and the theoretical RSD_{TR} predicted by the Horwitz model. The proposed *HorRat* acceptance interval is [0.5, 2.0] [26].

Over time, this criterion and its acceptance have acquired a semi-official character which seems to give a kind of award to a method or, on the contrary, reject it. For example, the FDA, the Codex Alimentarius, and the European Union have adopted Horwitz's model to decide whether a method can be used for official control purposes, as explained later in this chapter. It can be expected that this decision rule is somewhat arbitrary. As remembered, the model is empirical and has no physical nor physicochemical basis that could justify a formal relationship between concentration and measurement precision. From an analytical point of view, one given method may be highly reproducible while another may not, but the main point is that both can fulfill the objectives assigned and are suitable for a given purpose.

Figure 7.6 illustrates Horwitz's model on a logarithmic scale ranging from 1 to 10^{-12} kg/kg or from 1 kg/kg to 1 ppt, as recommended by the author. A few values have special meaning for the customary laboratory discussion: $10^{-2,}$ which stands for 1%, 10^{-6} for part per million or ppm, 10^{-9} for part per billion or ppb, and 10^{-14} for part per trillion or ppt.

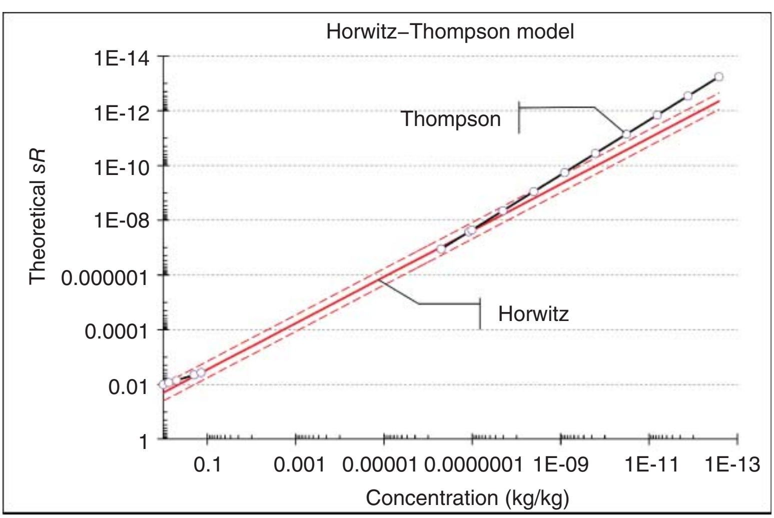

Figure 7.6 Horwitz (solid line) and Thompson (dashed line) models. Both axes have logarithmic scales. Two thin dotted lines outline the *HorRat*.

A simple calculation shows that for 1.0 ppm, the theoretical standard deviation of reproducibility should be 0.17 ppm. If, abusively, it was assimilated to a standard uncertainty, then it is necessary to apply a coverage factor of 2.0, as explained in Section 6.10, and the expanded uncertainty would be 0.34 ppm or in a relative form 34%. However, this reasoning is somewhat fallacious. The reproducibility of a method is not the uncertainty of measurement [17].

It should also be noted that for concentrations below 10^{-10} kg/kg, or 0.01 ppm, the RSD_{TR} is 50% or more. This means a dispersion of ±100% if the coverage factor of 2.0 is applied. At this concentration level, the values provided by the Horwitz model may become inapplicable for official control since they are between 0 and 2 ppm.

Let us consider a practical application. The Codex Alimentarius and FDA have set a maximum residue limit (MRL) for aflatoxin M1 in dairy products at 0.5 μg/kg, or 5×10^{-10} kg/kg [27]. For this concentration, the Horwitz model gives the theoretical value $RSD_{TR} = 54\%$. In terms of dispersion of the measurements, this would be about 100%, and the sample acceptance range [0.0, 1.0] 10^{-10} kg/kg. However, this reasoning is somewhat fallacious but underlines some pratical application problems. The reproducibility of a method is not the uncertainty of a measurement.

Another model has been proposed as a set of three equations depending on the concentration [28]. The Horwitz-Thompson model is illustrated in Figure 7.6 by the large, black line.

Concentration (kg/kg)	**Standard deviation of reproducibility**
$X < 1.2 \times 10^{-7}$	$s_{TR} = 0.22 \times X$
$1.2 \times 10^{-7} \leq X \leq 0.138$	$s_{TR} = 0.02 \times X^{0.845}$
$X > 0.138$	$s_{TR} = 0.01 \times X^{0.5}$

The main disadvantage of the Horwitz–Thompson model is to be obtained from a wide variety of measurement methods. To have more specific method-oriented model, the revised version of ISO 5725-2 for interlaboratory studies includes a set of *functional relationships* between concentration and observed s_R other than the Horwitz–Thompson model. In other words, empirical relationships between these two quantities. Four models, numbered from I to IV, are proposed to relate the standard deviation of reproducibility s_R to X, as summarized in Table 7.9.

These relationships allow more realistic models to be fitted and adapted to each analytical technique. Similar models describe the standard deviation of repeatability. Unfortunately, ISO 5725-2:2019 does not specify how to choose the *best* relationship; there is no acceptance or quality criterion, as explained by *HorRat*. A major part of regulatory guidelines uses percentages to define parameter acceptance limits. For this reason, the Horwitz model was quickly adopted and became a suitable reference for the FDA, the Codex Alimentarius or the European Union, and many others [24, 29, 30].

Besides the acceptance limits for precision, acceptance intervals for trueness were added, as illustrated in Table 7.10. This is a composite outline of several documents,

Table 7.9 Models relating s_R to Z proposed in ISO 5725-2:2019.

Model	Type	Model	
I	Linear through 0	$s_R = bZ$	(7.23)
II	Linear	$s_R = a + bZ$	(7.24)
III	Quadratic	$s_R^2 = a^2 + (bZ)^2$	(7.25)
IV	Power	$s_R = aZ^b$	(7.26)
	Linearized form	$\log(s_R) = \log(a) + b.\log(Z)$	

Table 7.10 Trueness and precision acceptance limits applied in several official regulatory guidelines.

		Recovery yield (%)		Reproducibility RSD (%)	
Concentration (kg/kg)	**Unit**	**Lower (%)**	**Upper (%)**	**Expected (%)**	**Horwitz (%)**
1	100%	98	102	1.3	2
10^{-1}	10%	98	102	1.9	3
10^{-2}	1%	97	103	2.7	4
10^{-3}	1‰	95	105	3.7	6
10^{-4}	100 ppm	90	107	5.3	8
10^{-5}	10 ppm	80	110	7.3	11
10^{-6}	1 ppm	80	110	11.0	16
10^{-7}	100 ppb	80	110	15.0	22
10^{-8}	10 ppb	60	115	21.0	32
10^{-9}	1 ppb	40	120	30.0	45

and more precise values can be found in official documents. The variation range of concentration is exceptionally large and expressed power of 10 for kg/kg, as in Figure 7.6. The second column gives the informal name of the concentration units often used in practice. Figure 7.7 is a graphical illustration of this table. The horizontal axis is generally expressed in logarithms to cover a broad range of concentrations. This indicates the leading role of concentration in the definition of relevant acceptance limits.

As explained in Section 4.1.1, the recovery yield is considered by regulators as the most convenient parameter to express trueness. It is noteworthy that all acceptance criteria are expressed as percentages. This practice is quite common in analytical sciences, but is not without drawbacks. Just like any ratio, it is a combination of two values. In the present context, both have the same units, resulting in a dimensionless parameter. Consequently, the initial scaling is lost, e.g. one does not know if the value is obtained at ppb or ppm level.

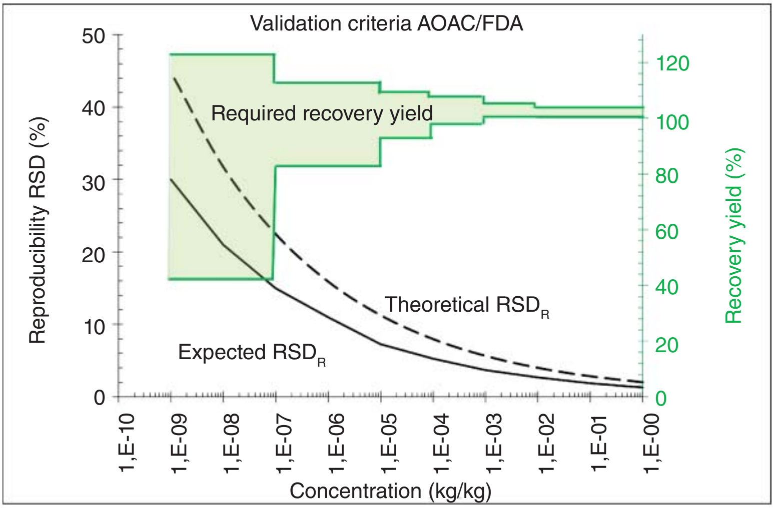

Figure 7.7 Precision and trueness acceptance criteria proposed in various official guidelines.

7.5.2 Fitting the Uncertainty Function

It was previously emphasized that the Horwitz and other models relating reproducibility and concentration cannot be directly used to predict MU at any concentration but only suggest that a functional relationship does exist between measurement dispersion and concentration. This relationship is hereafter referred to as the uncertainty function of measurement. Having such a function is interesting for the routine use of the method because it will allow us to associate an MU with any unknown sample. The only limitation is the concentration of the sample within the range where this function has been determined.

For example, if one relies on the data of an accuracy profile, the uncertainty function will only apply in the validated range. Table 7.11 brings together the main uncertainty functions encountered in the literature. They are presented in two forms deductible from each other, either the standard uncertainty $u(Z)$ or the relative uncertainty $UR\%$, respectively; this link is logical as $UR\% = u(Z)/X$. Except, in the case of the constant function, only two parameters, noted a and b, fully describ the uncertainty function of the method. This condensed form is very convenient when several MU estimates are calculated.

Both MU estimates, standard uncertainty and relative uncertainty, are computed from the THEOPHYLLINE dataset collected in Table 7.3. They are graphically illustrated for the standard uncertainty $u(Z)$ and for the relative uncertainty $UR\%(Z)$ in relation to the concentration Z in Figure 7.8a,b, respectively. They clearly illustrate the increase of variability of MU as the theophylline concentration decreases. In this example, a power function gives a good fitness of this variation in the validated domain.

To calculate the coefficients of a power function using Excel built-in functions, such as `LINEST`, `SLOPE`, or `INTERCEPT`, the simplest method is to log-transform

Table 7.11 Main uncertainty functions.

Type	Standard uncertainty	Relative uncertainty	
Constant	$u(Z) = a$	$UR\% = \left(\frac{2a}{Z}\right) \times 100$	(7.27)
Proportional	$u(Z) = bZ$	$UR\% = 2b \times 100$	(7.28)
Linear	$u(Z) = a + bZ$	$UR\% = \left(2b + \frac{2a}{Z}\right) \times 100$	(7.29)
Power[a)]	$u(Z) = aZ^b$	$UR\% = (2aZ^{b-1}) \times 100$	(7.30)
		$UR\% = (2aZ^{1+b})$	(7.31)

a) An explanation of these two equivalent formulas is given below.

concentration and uncertainty before the computation. This is possible with the `LN`, `LOG10`, or `LOG` functions. As shown in rows 7–9 of the Resource M worksheet, the most convenient transformation is the decimal logarithm `LOG10`. The antilog of the slope is easy to obtain, as illustrated in rows 12 and 15 of the worksheet.

Resource M Calculation of the coefficients of a power function (Excel).

	A	B	C	D	E	F	G	H
1	**Resource M: Calculation of the coefficients of a power function**							
2	Theophylline (µg/l)							
3	*Z*	0.05	0.1	0.5	1	2.5	10	
4	*u*(*Z*)	0.0117	0.0130	0.0350	0.0862	0.2749	0.5093	
5	*U*(*Z*)	0.0234	0.0260	0.0700	0.1724	0.5497	1.0186	
6	*UR*%(*Z*)	46.8%	26.0%	14.0%	17.2%	22.0%	10.2%	
7	Log10(*Z*)	−1.301	−1.000	−0.301	0.000	0.398	1.000	=LOG10(G3)
8	Log10(*u*(*Z*))	−1.932	−1.886	−1.456	−1.065	−0.561	−0.293	=LOG10(G4)
9	Log10(*UR*%)	−0.330	−0.585	−0.854	−0.764	−0.658	−0.992	=LOG10(G6)
10								
11	**Standard uncertainty function coefficients**							
12	Constant	0.0907	=10^INTERCEPT(B8:G8;B7:G7)					
13	Power	0.7780	=SLOPE(B8:G8;B7:G7)					
14	**Relative uncertainty function coefficients**				**Verification**			
15	Constant	0.1813	=10^INTERCEPT(B9:G9;B7:G7)			0.1813	=B12*2	
16	Power	−0.2220	=SLOPE(B9:G9;B7:G7)			−0.2220	=B13-1	

With the THEOPHYLLINE data (inverse-predicted by the quadratic WLS regression models), the uncertainty function coefficients are:

Parameters	*u*(*Z*)	*UR*%
Constant:	0.0907	0.1813
Power:	0.7780	−0.2220

These values are the same as those in Figure 7.8a,b, directly obtained with Microsoft Excel by adding a trendline with the option "Power" on the graphics. When fitting the relative uncertainty power function in Excel, the *UR*% values are